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**Macro-Micro Robotic System with a
Flexible Continuum Tip for Minimally
Invasive Surgery**

Interim Report and Project Plan

Abstract

This interim report and project plan is a Thesis A submission that lays the groundwork for improvements to a UR5e-based macro-micro robotic system with a flexible continuum tip for minimally invasive surgery (MIS). It consolidates relevant findings from prior work and available literature in relevant fields, and critically analyses how these prior works are related to this thesis. The review explores three technical threads relating to improvements of the UR5e-based system. Control improvements for teleoperation, tool exchange mechanisms for modularity, and safety and reliability considerations are reviewed, with a focus on identifying specific gaps that this thesis can address. The report finds clear research opportunities in these areas, specifically around smoothing for teleoperative control, integrated tool exchange validation, and comprehensive safety analysis.

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Nomenclature

A_c	=	tendon/cable cross-sectional area
D	=	backlash deadband
E	=	Young's modulus of tendon/cable material
M_{grasp}	=	estimated distal grasping torque
R_0	=	driving pulley radius
R_5	=	driven pulley radius
R_{eq}	=	equivalent-circle radius for tendon–sheath geometry
R_i	=	orientation matrix of segment i
T_i	=	homogeneous transform of segment i
$\Delta\theta_{out}$	=	output-side angular backlash/displacement loss
Ω	=	total cumulated curve angle of a tendon–sheath path
θ_{comp}	=	feedforward backlash compensation term
μ	=	friction coefficient (cable–pulley contact)
θ_i	=	local bending angle of segment i
α_i	=	local orientation offset of segment i
$\mathbf{p}$	=	rod centerline position in Cosserat formulation
$\mathbf{R}$	=	rod orientation in Cosserat formulation
$\mathbf{n}$	=	internal force in Cosserat formulation
$\mathbf{m}$	=	internal moment in Cosserat formulation
MIS	=	minimally invasive surgery

1 Introduction

1.1 Project Context and Problem Significance

This report aims to extend on previous a teleoperated macro–micro surgical robot prototype developed by 2023 UNSW thesis student Lachlan Scott [1]. Scott’s project established the viability of such a system, however he outlined potential areas of improvement, which this report addresses. Furthermore, new improvement and research possibilities have been identified, such as smoothing the motion of the robotic UR5e arm, the modularity of the system and thus ease of tool exchange, and the safety and reliability of the system for clinical use. Hence, this review of the literature and the thesis question that follows need to recognise this context with regards to Scott’s work. This report accordingly defines a focused research question and supporting project plan, and describes work undertaken to ensure feasibility and alignment with the identified knowledge gaps.

2 Literature Review

2.1 Minimally Invasive Surgery (MIS) and Robotic Systems

2.1.1 Historical Progression of Surgery and MIS Robotics

Technology and surgery are inseparable; every surgical advancement or care improvement has been enabled by the parallel development of instruments and tools. From anaesthesia to antiseptics, innovation has allowed surgeons to reduce risk and expand the scope of treatable conditions [2]. In the 19th century, the transition from candlelight to electric lighting allowed for practical internal visualisation, marking the beginning of a major paradigm shift in the surgical field [3]. Electricity also enabled development of various specialised surgical procedures, such as electrocautery for haemostasis and the use of electric drills for orthopaedic surgery [3]. Over time, the focus has shifted towards precision and dexterity, and techniques such as laparoscopy and endoscopy were developed at the beginning of the 20th century, resulting in some of the first minimally invasive surgeries (MIS) [4]. Laparoscopic surgery is a core MIS approach in which instruments are inserted through small abdominal incisions using trocar access ports, rather than a large open incision [6]. Holistically, MIS denotes procedures that use these reduced-access pathways to lower tissue trauma and recovery burden relative to open surgery. This trend towards less invasive procedures has continued, and revolutionary robotics like the da Vinci Surgical System became widely adopted in the 1990’s and 2000’s [5], however there are still challenges to overcome.

2.1.2 Macro–Micro Architecture for Surgical Robotics

The current aspiration of modern surgical robotics is achieving human-level dexterity, accuracy, and force control in increasingly small and complex anatomical spaces with limited access and vision. One key technological direction enabling this transition is MIS tooling with a macro–micro architecture. This terminology refers to a robotic system where a larger, more rigid macro manipulator provides positioning and stability, while a smaller, more flexible micro manipulator is attached to the distal end of the macro arm, providing finer control in confined spaces [1].

The macro unit provides large-workspace positioning and efficiency; however, its larger structure can be prone to deformation and vibration. In contrast, the micro manipulation unit provides the distal dexterity required for manipulation in confined anatomy, where surgical targets often require end-effectors in the 3–4 mm class for practical access through narrow channels and natural orifices.

An example of this macro–micro setup is shown in Fig. 1, where a UR5e macro robotic arm is integrated with a tendon-driven continuum micro-manipulator [1].

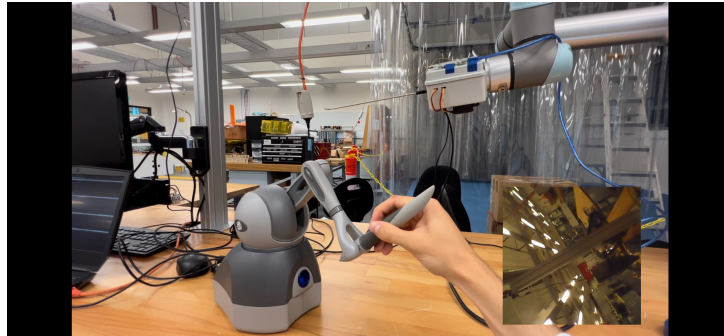


Figure 1: Macro–micro teleoperation system combining a UR5e robotic arm with a continuum-tip manipulator [1].

In current snake-like manipulator literature, this micro stage is typically tendon or cable driven to achieve controlled high curvature and orientation control, but the performance relies on the quality of motion transmission, sensing, and control architecture involved [7]. Regarding macro–micro robotics, this local dexterity complements the macro arm’s gross positioning capability and could isolate the endpoint from macro-stage vibration through coordinated control [1]. A system-level representation of the implementation of this concept is shown in Fig. 2, illuminating the component integration.

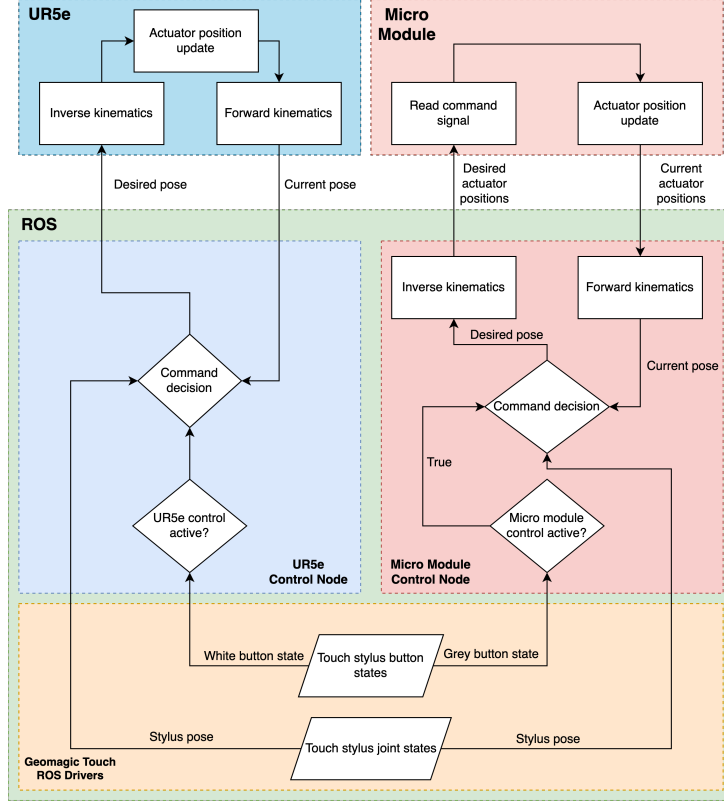


Figure 2: System architecture of a UR5e-based macro-micro surgical manipulator [1].

2.1.3 Continuum Manipulators for MIS

Continuum tips are a type of hyper-redundant end-effector that can bend continuously along their length. Their high dexterity and slender geometry allow them to navigate tight, curved pathways such as natural orifices and small incisions, where traditional rigid tools may not fit. Tool curvature is generated through differential tension about a central backbone, resulting in multi-DOF distal steering, making these tips suited to macro-micro architectures [7]. For example, the tendon driven continuum tip in Fig. 3 is integrated with a Raven II surgical robot for teleoperation [8].

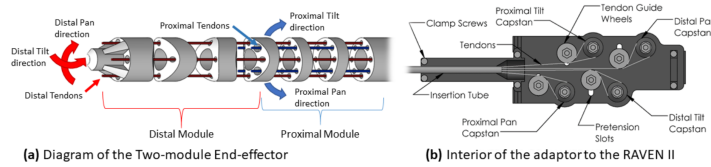


Figure 3: SnakeRaven macro-micro surgical manipulator concept [8].

2.1.4 Cost and Accessibility Considerations

Growing demand and an expanding surgical robotics market has provoked ongoing debate over whether the high cost of integrated platforms such as the da Vinci system limits accessibility. Acquisition costs have been reported up to \$4 million AUD, with additional per-procedure costs of approximately \$1000–\$5000 [9]. Despite clear clinical benefits, providers must still evaluate cost-effectiveness, especially in low and middle-income environments where there are more adoption barriers than the initial cost. Recurring maintenance, disposable instrument costs, and sustained training requirements can prevent usage even after installation [10].

Systemic constraints can also arise from limited infrastructure, workforce training gaps, and the need to prioritise basic surgical capacity over advanced robotic systems [10]. These factors collectively highlight a significant translational gap between technological capability and real-world clinical deployment. This literature indicates that scalable, cost-effective robotic solutions remain underdeveloped, and thus there is a need for affordable macro–micro systems that combine cheaper hardware with clinically useful teleoperation performance. This aligns with broader advocacy of off-the-shelf hardware and open source software frameworks to reduce cost yet maintain the precision and safety requirements of minimally invasive surgery [1].

2.1.5 UR5e-Based Systems and Identified Limitations

The UR5e is a common, economical robotic manipulator that can serve as the macro stage in macro–micro surgical architectures, with reported pricing around \$50,000 to \$60,000 AUD [11]. Its lower cost and flexibility makes it suitable for integration with custom micro manipulation components. The feasibility of this architecture has been demonstrated in prior thesis work by Scott [1], which implemented a proof-of-concept teleoperation pipeline using open-source ROS2. Within this framework, a tendon-driven continuum micro-manipulator was integrated with the macro arm and operated via a haptic interface, following the snake-like design approaches commonly reported in the literature.

Preliminary validation using electromagnetic tracking demonstrated close pose mirroring under teleoperation, with reported average absolute position errors of 0.0272 mm and 0.0161 mm across trial groups. These results indicate that high-precision control is achievable within such macro–micro configurations.

However, several practical engineering limitations remain. In particular, translational commands at the macro stage were observed to induce vibration in the integrated system,

negatively affecting motion stability and repeatability. Additionally, the current modular workflow requires manual exchange of tendon-driven end-effectors, resulting in increased setup time and reduced operational efficiency.

These limitations highlight a key gap in existing UR5e-based macro–micro systems: while feasibility and precision have been demonstrated, improvements in vibration suppression, system robustness, and workflow integration are required to meet clinical expectations for safety and usability in real-world operating environments [1], [7], [8].

2.2 Teleoperation and Control in Surgical Robotics

2.2.1 Modelling Frameworks for Continuum Surgical Robotics

To situate control and modularity claims on a consistent mathematical basis, three modelling layers from continuum-robotics literature are directly relevant to this project: kinematic representations, reduced-order curvature models, and mechanics-based formulations. At the kinematic level, each segment transform can be expressed as

$$T_i = \begin{bmatrix} R_i & p_i & \mathbf{0}^\top & 1 \end{bmatrix}, \quad (1)$$

which maps segment states to distal pose and forms the baseline representation used across continuum and hyper-redundant robotic systems [12], [13]. This formulation is computationally efficient and well suited to real-time teleoperation, but does not explicitly capture load-dependent deformation.

For tendon-driven continuum manipulators, piecewise-constant-curvature (PCC) models provide a practical actuator-to-workspace mapping:

$$T_i^{i+1} = \text{rot}(Z_i, \alpha_i), \text{trans}(Z_i, H_i), \text{trans}(X_i, W_i), \text{rot}(Y_i, \theta_i), \text{rot}(Z_i, -\alpha_i), \quad (2)$$

which underpins many high-DOF modelling and calibration approaches [13], [14]. PCC models strike a balance between computational efficiency and geometric fidelity, making them widely adopted in teleoperated continuum systems.

When compliance, external loading, and routing-dependent behaviour become dominant, mechanics-based approaches such as Cosserat-rod models provide a more complete representation of continuum dynamics. These models capture distributed force and moment balance along the manipulator backbone, enabling analysis of tendon friction, hysteresis, and deformation under load **janabi2021continuum**.

In critical terms, the literature indicates that no single modelling layer is sufficient for

workflow-ready surgical performance: kinematic models are fast but approximate, while full mechanics-based models are more expressive but computationally intensive and harder to identify in real time. This trade-off directly motivates hybrid modelling and compensation strategies in modern teleoperation systems.

Within this control context, the relevant evidence can be interpreted across five linked layers: tremor suppression, latency compensation, robust state estimation, predictive bilateral control, and intent-aware shared autonomy. This layered interpretation is summarised in Fig. 4.

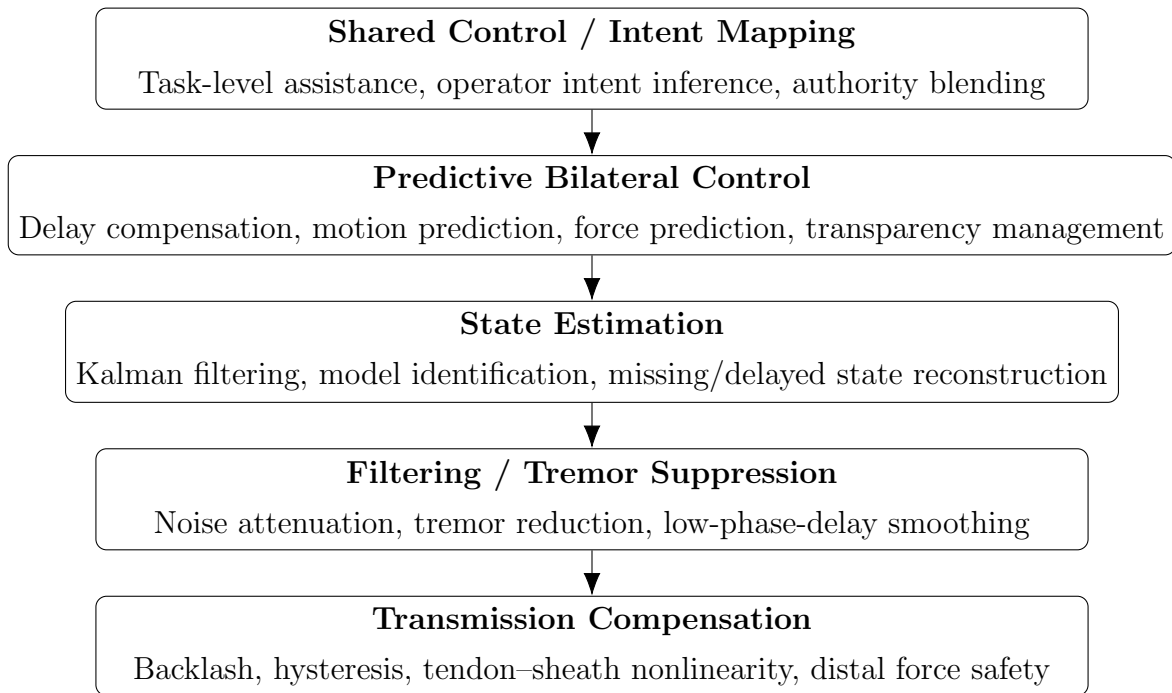


Figure 4: Linked control layers relevant to teleoperation in macro-micro surgical robotics. The literature reviewed in this subsection can be interpreted across these interacting layers, from low-level transmission effects to high-level shared control.

2.2.2 Tremor Suppression and Delay-Robust Interaction

Human physiological tremor is a central control challenge in robot-assisted MIS because involuntary operator motion is transmitted through the teleoperation loop and can appear as vibration at the surgical instrument tip, degrading precision during fine tissue manipulation, suturing, and needle handling. Sang, Yang, Liu, *et al.* [15] report that tremor suppression methods must balance two competing requirements: strong attenuation of involuntary motion and minimal phase delay, as latency in the command pathway can directly reduce teleoperation quality. Their zero-phase adaptive fuzzy Kalman filter (ZPAFKF) combines a zero-phase prefilter with adaptive Kalman estimation to improve

tremor estimation and compensation in real time, and experimental master–slave results show lower error and reduced tip vibration relative to standard filtering baselines. This is directly relevant to the present thesis because it frames tremor control not as a standalone signal-processing problem, but as a coupled teleoperation and tracking problem in which compensation quality, delay, and manipulator dynamics must be considered together [15].

System latency is a second critical bottleneck in bilateral teleoperation, particularly when haptic feedback is included. Batty, Ehrampoosh, Shirinzadeh, *et al.* [16] show that communication delay creates a fundamental transparency–stability trade-off: increasing direct force correspondence can degrade stability, while aggressive stabilisation can mute the fidelity of operator feedback. In response, they implement an environment-estimation and force-prediction architecture, where slave-side contact parameters are estimated online (EWRLS) and master feedback is generated through a predicted virtual environment rather than delayed direct force reflection. Their human-in-the-loop palpation experiments demonstrate that this predictive approach substantially reduces force prediction error (with the Hunt–Crossley model outperforming Kelvin–Voigt and direct force feedback), and improves teleoperation transparency under delay. A key implication in the literature is that latency management should be treated as a control-architecture problem, not solely a communications problem, and that model quality strongly influences the practical quality of teleoperated manipulation [16]. Effective solutions must therefore balance filtering, prediction, and system dynamics to maintain both stability and responsiveness.

2.2.3 State Estimation and Predictive Bilateral Architectures

Filtering is therefore not optional in surgical teleoperation; it is a foundational requirement for maintaining stable and trustworthy state estimates when measurements are noisy and network transport is imperfect. Lashari, Batayneh, and Khokhar [17] show that a Kalman filter paired with MOESP-identified state-space models can estimate patient-side manipulator position with high robustness under delay, jitter, and packet loss, while remaining computationally lightweight for real-time use. This is especially relevant when exact plant dynamics are not available a priori, because MOESP builds the model directly from surgical kinematic data (JIGSAWS), allowing filtering performance to be grounded in observed system behaviour rather than idealised assumptions. In practical terms, filters are needed to smooth stochastic disturbances, reconstruct missing or delayed information, and preserve control continuity so that downstream motion control and haptic/visual feedback remain clinically usable. A key limitation, however, is that the reported performance is primarily based on dataset-driven modelling and simulated network impairments, so in vivo generalisability and cross-platform transfer still require further validation [17].

Predictive bilateral control is also emerging as a practical strategy for continuum and snake-like teleoperation under delay. Ebrahimian, Pourmokhtari, Ghiyasi, *et al.* [18] propose an online architecture that combines model-mediated force prediction on the master side with ANN-based operator motion prediction on the slave side, with the explicit goal of reducing perceived communication lag while preserving position synchronisation and haptic transparency. This is highly relevant to the macro–micro literature because it addresses the same core bottleneck as latency and filtering studies, but at the controller-architecture level: delay compensation is handled through coordinated forward and backward prediction, rather than through one-sided compensation only. Their simulation results show strong delay bypass for known constant delays in snake-like robot teleoperation, but a key limitation is that performance is demonstrated mainly in simulation with structured assumptions (e.g., known constant delay and constrained robot DOFs), which may not fully capture variable-delay clinical networks or higher-complexity surgical interaction dynamics [18]. Together, these approaches highlight that robust teleoperation under delay requires both reliable state estimation and predictive control, reinforcing the need for integrated architectures rather than isolated solutions.

2.2.4 Shared Control, Immersion, and Haptic Force Awareness

Beyond tremor and delay, user intent mapping is a central teleoperation design problem in shared control. Bowman, Zhang, and Zhang [19] argue that direct motion mapping alone is insufficient for robust telemanipulation when the human and robot hands are physically dissimilar, because low-level geometric tracking does not necessarily preserve high-level task success. Their intent-based task-oriented shared control framework instead blends operator-following behaviour with autonomous task assistance using inferred intent probabilities and adaptive arbitration weights. Critically, this contribution is not just another blending policy: it formalises how authority should shift based on hand-structure discrepancy and task constraints, which is directly relevant to macro–micro MIS systems where distal tools cannot perfectly mimic surgeon hand kinematics. This supports the broader literature position that intuitive teleoperation should be evaluated by both task completion and perceived control, rather than endpoint tracking error alone [19].

However, an important limitation is that the reported validation is constrained to controlled grasping tasks and specific intent models, so transferability to highly dynamic, deformable, and safety-critical in vivo surgical contexts remains an open question.

Recent review-level synthesis also indicates that immersive visualisation and haptic augmentation are transitioning from laboratory prototypes toward clinically relevant teleoperation workflows [20]. Across the studies summarised, VR-based viewpoint control and

stereoscopic scene rendering improve situational awareness in tasks where fixed-camera teleoperation typically underperforms. Representative evidence includes an increase in pick-and-place success from 62.5% to 90.0% in a viewpoint-independent VR interface trial, and reduced operator workload and task time in mixed-reality assisted manipulation experiments [20].

For MIS relevance, the strongest transferable point is not immersion alone but the interaction between perception quality and force regulation. Bogue [20] report that force-feedback-enabled surgical platforms are now appearing in commercial systems, including preclinical da Vinci 5 results where surgeons applied 43% less tissue force with haptic-enabled instruments, and Saroa trial results showing mean grasp-force reduction from 2.14 N to 0.63 N in a delicate transfer task. These results support including haptic and force-awareness as a safety-linked teleoperation design variable in this thesis, rather than treating it as an optional interface enhancement.

The key limitation is evidentiary heterogeneity: the article aggregates diverse platforms, task types, and evaluation protocols, so direct cross-study performance comparison is limited. For thesis use, this source is therefore most valuable as trend evidence motivating evaluation criteria such as situational awareness, workload, and applied-force moderation, while primary validation claims should still rely on the underlying experimental studies.

2.2.5 Transmission Nonlinearity and Distal Force Safety

Transmission backlash in cable-pulley drives is another directly relevant control bottleneck for laparoscopic end-effectors because direction reversals create delay and transition phases in which distal motion and torque do not track proximal commands linearly. Zou, Li, and Huang [21] model this explicitly and use a feedforward compensation term linked to output-side backlash, $\theta_{comp} = \Delta\theta_{out}, R_5/R_0$, together with grasping torque estimation and saturation limiting, $M_{grasp} = M_{out} - M_{fric}$, to reduce tracking error and limit unsafe distal force. Their results indicate meaningful improvements in position and force control for cable-pulley laparoscopic instruments: reported distal-grasper position error is reduced to approximately 1.03° mean (3.49° max), and grasping torque tracking error is reported below 0.025 N·m in compensated instrument tests. Nonetheless, residual error around direction reversal remains, highlighting that compensation quality still depends on parameter identification and friction-model fidelity [21].

Related tendon–sheath evidence further strengthens this point in shape-variable conditions. Hong, Hong, Kim, *et al.* [22] model backlash-like hysteresis in tendon–sheath pairs using an equivalent-circle representation, where deadband is linked to master/slave deformation mismatch ($D = \delta_{TSM(master)} - \delta_{TSM(slave)}$) and then compensated by feedforward

updated from proximal calibration without distal sensing. Their results show RMSE reduction from 1.31 mm to 0.19 mm on average (85.5% reduction). This is highly relevant to laparoscopic routing uncertainty; however, the approach still depends on reliable calibration and assumptions about unchanged geometric conditions during each compensation interval [22].

Collectively, these studies highlight that transmission nonlinearity remains a critical limitation in tendon-driven surgical manipulators, particularly under variable routing and loading conditions. While feedforward compensation can significantly reduce tracking and force errors, performance remains dependent on accurate parameter identification and stable system geometry.

These findings establish clear performance requirements for low-cost macro–micro platforms and provide a direct conceptual bridge to modular continuum hardware discussions. The relationship between these control problems within the overall teleoperation loop is illustrated in Fig. 5.

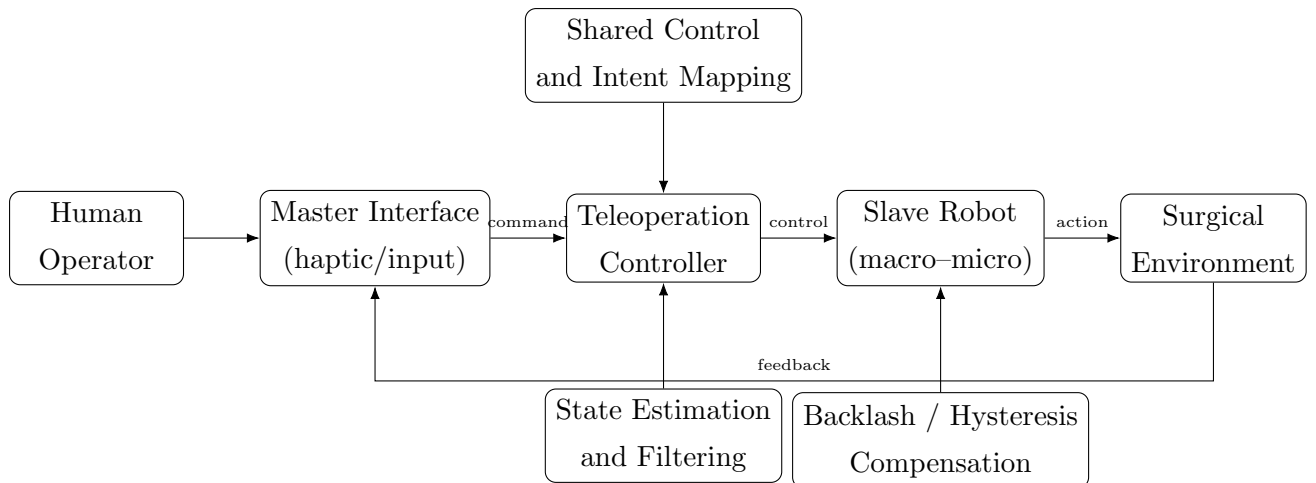


Figure 5: Simplified teleoperation architecture for macro–micro surgical robotics. The reviewed literature addresses different parts of this loop, including state estimation, shared control, predictive control, and transmission compensation.

2.3 Modularity and Tendon-Driven Continuum Manipulators

2.3.1 Monolithic vs Modular Distal Architectures

In this review, a *monolithic* distal architecture is defined as a single integrated tool assembly in which changing function requires major teardown or full distal module replacement, rather than rapid local exchange. This framing is consistent with the architecture-level distinction between integrated and modular robotic platforms reported by Cowan, Gomes,

Morris, *et al.* [23].

The practical consequences are workflow-critical, including longer swap times, re-tensioning and re-calibration burden after replacement, interruption of the operative sequence, and reduced intraoperative adaptability when task requirements change mid-procedure. These constraints align with broader integration-friction evidence in modular end-effector software and hardware pipelines [24], where even technically feasible modularity often carries non-trivial changeover overhead.

Internal platform observations in prior local UR5e-continuum implementations also indicate that end-effector replacement is time-consuming and introduces practical workflow friction in achieving repeatable setup after each swap [1]. Together, the literature and local prior-work evidence indicate that architectural modularity alone is insufficient unless exchange time and post-swap recalibration costs are explicitly addressed.

2.3.2 Tool Exchange Mechanisms

Tool exchange is necessary because laparoscopic workflows require multiple functional instruments within a single procedure, while only a limited subset can be mounted simultaneously on the robotic arm [6]. This creates a recurrent intraoperative transition task that can interrupt surgical flow and, when assistant-dependent, introduces operator–assistant coordination variability.

2.3.3 Rail-Guided and Cartridge-Based Exchange

At present, the strongest direct evidence remains with rail-guided, cartridge-like exchange approaches. Ko, Kim, Lim, *et al.* [6] analyse current sliding exchange practice and report an automated replacement architecture designed to preserve trocar and remote centre of motion (RCM) constraints via linear motion. Their prototype-level results report full exchange in 15 s, reduced from an initial 27 s cycle, and a total device mass of 3.36 kg (below their 4 kg design target), indicating feasible automation under laparoscopic geometric constraints.

2.3.4 Reconfigurable and Multifunctional Instruments

Alternative strategies aim to reduce exchange frequency rather than improve exchange interfaces directly. Mocellin, Pagliarani, Gamberini, *et al.* [25] present OriGrasp as a reconfigurable multifunctional end-effector (introducer mode and compliant grasper mode), validated through dVRK integration and tissue-interaction tests. Reported performance includes approximately 4 N pinch force and bowel pulling/lifting forces in literature-consistent atraumatic ranges. This supports multifunctionality as a useful partial mit-

igation for selected abdominal tasks. However, current evidence remains task-specific and does not demonstrate that a single reconfigurable device can replace the broader instrument set required across heterogeneous procedures.

2.3.5 Limitations of Alternative Exchange Mechanisms

Mechanism diversity beyond rail-guided exchange remains under-validated in the reviewed evidence base. Direct surgical evidence is limited for bayonet quarter-turn, cam-latch, and cartridge-head variants beyond linear sliding, as well as kinematic-coupling-assisted docking with active preload. Reported gaps are not purely mechanical: tolerance sensitivity, sterilisation robustness, wear-dependent repeatability drift, and post-reconnection re-zeroing burden all remain insufficiently quantified across repeated swap cycles.

2.3.6 Post-Swap Control Recovery and Tension Management

Control evidence indicates that docking success alone is insufficient for workflow-ready exchange. Cable-pulley and tendon–sheath studies report backlash, hysteresis, and force-estimation sensitivity to pretension, friction, and routing geometry [21], [22]. In particular, Hong, Hong, Kim, *et al.* [22] show geometry-dependent deadband behaviour and report RMSE reduction from 1.31 mm to 0.19 mm with feedforward compensation after calibration, implying that post-swap recovery may require both tension re-establishment and compensation-state updates when routing conditions change.

For post-swap tension management specifically, Singh and Khatait [26] provide relevant but indirect evidence via an RCM tension-adjustment mechanism enabling independent cable pretension tuning with invariant routed tendon length. The key limitation is translational scope: this is not a direct quick-change end-effector interface study and does not quantify repeated dock–undock surgical workflow performance.

A comparative summary of the main tool-exchange strategy families discussed in the literature is provided in Table 1.

Criterion	Rail-guided sliding	Quick-lock / bayonet	Reconfigurable tool
Mechanism concept	Linear cartridge-style or rail-guided insertion and removal	Rotational locking, latch-based docking, or compact quick-coupling interface	Single distal tool changes functional mode instead of being physically exchanged
Workflow benefit	Structured and repeatable alignment during exchange; compatible with trocar / RCM constraints	Potentially compact and rapid attachment with fewer gross insertion steps	Reduces exchange frequency by combining multiple functions into one tool
Main limitation	Still requires reliable docking, reconnection, and post-swap recovery	Direct evidence for repeatability, sterilisation robustness, and re-zeroing remains limited	Typically validated only for narrow task classes; unlikely to replace the full instrument set
Evidence base	Strongest direct exchange evidence in the reviewed literature	Sparse direct surgical validation and weak comparative evidence	Promising prototype/task-specific evidence, but limited generality across procedures
Validation maturity	High	Low	Moderate

Table 1: Comparison of tool-exchange strategy families relevant to modular tendon-driven surgical manipulators

2.3.7 Critical Assessment and Identified Knowledge Gap

The literature supports a clear hierarchy of maturity: (i) strongest evidence for linear, rail-guided exchange, (ii) promising but task-limited multifunctional substitution, and (iii) weak comparative evidence for alternative docking and locking mechanisms.

The unresolved research gap is therefore the integrated validation of *mechanism and control recovery* following repeated tool exchanges. In particular, exchange time, docking repeatability, post-swap backlash and deadband behaviour, retensioning effort, and distal force-safety consistency must be quantified together under clinically realistic routing and sterilisation conditions.

2.4 System Safety and Reliability Considerations

2.4.1 Safety Constraint Model and Hazard Taxonomy

Safety in macro–micro MIS is treated here as constrained execution under mechanical, control, and workflow limits. In minimally invasive surgery, tools are introduced through a *trocar*, a fixed entry port that enforces a pivoting constraint on the instrument shaft. This is commonly modelled as a remote centre of motion (RCM) constraint, requiring zero velocity at the entry point [12], [13].

At the kinematic level, trocar (RCM) constraint enforcement can be represented by

$$\mathbf{v} * RCM = J * RCM \dot{\mathbf{q}} = \mathbf{0}, \quad (3)$$

and in constrained form

$$\dot{\mathbf{x}}_e = J_c \dot{\mathbf{q}} * I, \quad J_c = J_e T \left[I_n - J * II^{-1} J_I \right], \quad (4)$$

which enforces motion consistency under kinematic constraints [13].

For moving trocar conditions, such as soft-tissue interaction or external disturbance, the constraint can be extended to

$$\dot{\mathbf{q}} * II = J * II^{-1} (\mathbf{v}_{trc} - J_I \dot{\mathbf{q}}_I), \quad (5)$$

preserving explicit constraint compensation under disturbance.

These equations define one hazard layer (constraint violation), but safety risk in this thesis is broader and includes distal force overshoot, device malfunction, post-swap state mismatch, delayed recovery from recoverable faults, and degradation from repeated re-processing.

2.4.2 Error Detection and Reporting Architecture

Rajih, Tholomier, Cormier, *et al.* [27] provide useful evidence on practical malfunction detection, as errors were captured through da Vinci Si event logs, time-stamped, and classified as recoverable versus unrecoverable across 1228 procedures. This is important for this review because safety analysis depends on what is actually detected and recorded, not only what fails physically.

Their workflow separates events by timing (pre-anesthesia setup, intraoperative, and post-undocking) and by subsystem category, creating a useful template for future macro–micro

logging: event source, recoverability, corrective action, and impact on operative flow [27].

2.4.3 Malfunction Frequency and Immediate Clinical Consequence

2.4.4 Malfunction Frequency and Clinical Consequence

Across the same cohort, Rajih, Tholomier, Cormier, *et al.* [27] report overall malfunctions in 4.97% (61/1228) of cases, with recoverable errors dominant (46/61). The most common event type was arm pressure/output limit faults (2.04%, 25/1228), followed by communication-related errors (1.06%, 13/1228), failed encoder errors (0.57%, 7/1228), illuminator faults (0.33%, 4/1228), and smaller electronic categories.

In immediate outcome terms, they report no malfunction-driven case abortion, no malfunction-driven conversion, no injury, and no death, with one delayed case due to a battery-related error. This supports their conclusion that malfunctions are not rare but are typically low-consequence when recovery pathways are present. However, this does not eliminate system-level risk, as consequence severity depends on context, detection latency, and available technical support [27].

To maintain explicit risk representation, this review uses

$$\hat{p} * i = \frac{N_i}{N * tot}, \quad RPN_i = S_i O_i D_i, \quad (6)$$

where $\hat{p}_i$ denotes category occurrence probability and RPN_i is a risk-priority metric based on severity (S_i), occurrence (O_i), and detectability (D_i), with confidence caveats for single-centre and voluntary-reporting datasets.

2.4.5 Preventable Versus Non-Preventable Faults

The Canadian dataset helps separate preventable execution faults from non-preventable electronic anomalies. Rajih, Tholomier, Cormier, *et al.* [27] identify collision and jerky-motion-related events as a major recoverable class and explicitly link their frequency to team learning curve and technique. In contrast, communication-board anomalies are treated as less predictable and often require manufacturer-mediated service.

This distinction is directly relevant for thesis design because mitigation strategies differ: port and arm planning, as well as operator training, can reduce preventable faults, whereas non-preventable faults require rapid diagnosis, robust recoverability logic, maintenance policy, and fallback workflow design [27].

2.4.6 Lifecycle Reliability: Reconfiguration, Wear, and Reprocessing

Short-horizon controller performance remains necessary but insufficient for comprehensive safety claims. Sadeghian, Zokaei, and Hadian Jazi [28] demonstrate feasible constrained tracking performance (including experimental task-space error on the order of ± 0.4 , mm), but this does not address prolonged workflow stress or repeated reconfiguration effects.

For contamination risk, Saito, Yasuhara, Murakoshi, *et al.* [29] show that residual protein levels remain substantially higher in robotic instruments than in conventional instruments after in-house cleaning (robotic: 650, 550, 530 $\mu\text{g}/\text{instrument}$ vs. ordinary: 16, 17, 17 $\mu\text{g}/\text{instrument}$; $P < .0001$), with an estimated total residual burden of approximately $2.8 \times 10^3, \mu\text{g}$ versus 98 μg . The practical implication is that sterilisation and reprocessing must be treated as tracked reliability variables when repeated tool exchange and reuse are core to platform operation.

2.4.7 Integrated Safety Gap and Evaluation Endpoints

The supported safety gap is not the absence of individual solutions; rather, it is the lack of integrated validation across *constraint control*, *malfunction detection and recovery*, and *lifecycle reprocessing reliability* within a single coherent macro–micro workflow.

Accordingly, later chapters should report unified evaluation endpoints rather than isolated metrics. These include trocar-point (RCM) error bounds, malfunction detection-to-recovery time, post-swap recovery time, residual deadband and backlash after reconfiguration, distal-force limit adherence, repeatability across repeated cycles, and post-reprocessing contamination bounds.

2.5 Synthesis of Literature and Knowledge Gaps

The reviewed literature establishes a coherent but incomplete technical chain for macro–micro MIS systems. Continuum and tendon-driven hardware studies support distal dexterity in constrained anatomy, while teleoperation and control studies show that tremor, delay, estimation uncertainty, and intent mapping are jointly performance-critical. Safety studies demonstrate that malfunction events are generally manageable but not negligible, and accessibility studies introduce a practical constraint: performance must be achieved under realistic cost, maintenance, and workforce limits, not only in high-resource laboratory settings.

Critically, these bodies of work are usually validated in isolation. Kinematic and compensation studies often assume controlled routing and loading conditions; teleoperation studies typically assume simplified communication conditions or bounded tasks; malfunction

studies are strong in event reporting but weaker in mechanism-level causal interpretation; and lifecycle reprocessing studies quantify residual burden but are rarely connected to control-state recovery following repeated instrument exchange. This fragmentation limits the explanatory strength of any single study when the research question concerns end-to-end workflow reliability.

In relation to this project, the conceptual relationship is direct. The UR5e macro stage provides accessible gross positioning, while the continuum distal stage provides local dexterity; however, the interface between them introduces coupled failure modes. These include vibration transmission from the macro arm, backlash and hysteresis in tendon-driven actuation, post-swap recalibration burden following tool exchange, and safety assurance under disturbance. The research problem is therefore not limited to improving a single controller or mechanism, but to determining whether a low-cost macro–micro architecture can maintain stable, smooth, and safe operation when control, mechanics, and reconfiguration are exercised together.

Based on this synthesis, the principal knowledge gaps are:

- **Macro-stage vibration and smoothing gap:** limited evidence on how macro-arm dynamics (e.g. UR5e vibration and compliance) propagate to distal motion, and how filtering, prediction, and control strategies can achieve smooth, delay-robust teleoperation without degrading responsiveness.
- **Modular exchange and recovery gap:** insufficient validation of rapid, repeatable end-effector exchange mechanisms that minimise downtime while maintaining consistent calibration, tensioning, and transmission behaviour across repeated swaps.
- **Integrated safety-in-operation gap:** limited evidence linking constraint enforcement, fault detection and recovery, and force regulation to task-level safety outcomes under realistic operating conditions, including disturbance and operator variability.
- **System-level validation gap:** a lack of unified experimental frameworks that evaluate control performance, mechanical reliability, and reconfiguration effects together, rather than in isolation, using clinically meaningful and repeatable metrics.

In this context, the thesis contribution is positioned as a systems-level evaluation: to design and assess a UR5e-based macro–micro platform using unified endpoints for constraint adherence, teleoperation smoothness and robustness, post-swap recovery, and safety-relevant performance, rather than reporting isolated improvements in individual subsystems.

2.5.1 Remaining Evidence Gaps and Future Directions

The literature can be further strengthened by recent work in several areas:

- experimental studies on vibration propagation and smoothing strategies in macro–micro teleoperation systems,
- variable-delay and packet-loss-robust bilateral and shared control in human-in-the-loop settings,
- safety assurance frameworks linking fault detection, bounded motion, and force-limited behaviour to clinical outcomes,
- quantitative evaluation of modular tool-exchange repeatability and calibration drift under repeated use,
- lifecycle reliability of tendon-driven instruments under repeated use and sterilisation.

3 Research Question and Project Plan

3.1 Research Question

Based on the reviewed literature, this thesis addresses the following research question:

Can a UR5e-based macro–micro robotic system achieve measurable improvements in teleoperation smoothness, repeatable end-effector exchange, and perform with safety constraints under controlled laboratory conditions?

It is hypothesised that applying smoothing control, tool exchange capability, and integrated system evaluation will reduce vibration, improve swap consistency and speed, and maintain safe operation within defined bounds. While macro–micro architectures have demonstrated high positional accuracy and feasibility [1], they remain affected by macro-stage vibration, manual and time-consuming end-effector exchange, and fragmented approaches to safety evaluation. The literature further shows that teleoperation performance, modularity, and safety are typically studied in isolation, limiting the ability to assess overall workflow reliability. Accordingly, this thesis focuses on improving and testing the system at a *systems level*, rather than proposing isolated algorithmic or mechanical improvements.

3.2 Project Plan

The project is organised as a progressive experimental workflow, in which teleoperation control, modular exchange, and safety are evaluated sequentially and then together. These are not independent tasks, rather they comprise a single experiment. Performance will be evaluated using quantitative metrics including motion smoothness, effector swap time and accuracy, and adherence to constraint and safety bounds.

With regards to teleoprative control, prior studies show that tremor, delay, and vibration transmission significantly affect precision and usability [15], [16]. In this project, effort is therefore directed towards establishing a stable baseline system and implementing a practical smoothing strategy that improves motion quality without introducing excessive latency. This work also serves as the foundation for all subsequent evaluation, as reliable teleoperation is required before modularity and safety can be meaningfully assessed.

The second component focuses on modular end-effector exchange. While tool exchange is necessary in laparoscopic workflows, the literature highlights a lack of validation of post-swap behaviour, particularly with respect to calibration, tensioning, and transmission effects [6], [22]. Rather than developing a complex new mechanism, this project prioritises a repeatable exchange and recovery workflow, allowing the impact of swapping on system performance to be measured directly. This aligns with the identified gap that mechanism design alone is insufficient without evaluating post-swap recovery.

The third component addresses safety in an operational context. Existing studies demonstrate that robotic malfunctions are not uncommon but are typically recoverable when appropriate detection and response mechanisms are present [27]. However, these are rarely linked to control behaviour or modular reconfiguration. This project therefore evaluates safety using integrated metrics, including constraint adherence, recovery time, and repeatability across operation cycles, rather than relying on isolated fault statistics.

The scope of the project is constrained by the available time (approximately 200 hours across Terms 2 and 3) and the use of an existing UR5e-based platform. As such, the focus is on implementing and validating representative solutions in each area, rather than developing complete new hardware or control frameworks. This ensures that the project remains feasible while still addressing the key gaps identified in the literature. The project relies on access to the UR5e platform, ROS2-based control infrastructure, and completion of required laboratory training and coursework.

A detailed timeline is provided in Fig. 6. The plan is organised across two terms, with Term 2 focused on system setup and teleoperation control, and Term 3 focused on modular

exchange evaluation, safety testing, and integrated experiments. The Gantt chart captures the specific task breakdown, dependencies, and time allocation.

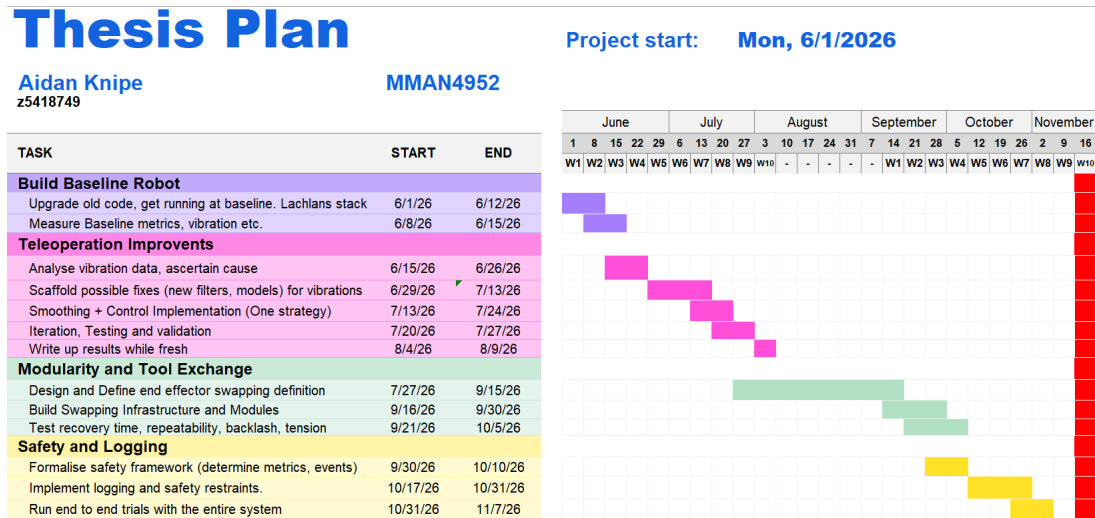


Figure 6: Project timeline and task allocation across Terms 2 and 3.

To support planning and identify potential risks, a structured mind map has been developed (Fig. 7). This includes task dependencies, alternative approaches, and anticipated failure points. In particular, it highlights areas where scope may need to be reduced, such as simplifying the exchange mechanism or limiting the complexity of control strategies, ensuring that meaningful results can still be obtained within the available timeframe. Key contingencies include simplifying the control strategy if integration proves difficult, and validating a manual but repeatable exchange workflow if hardware modifications are not feasible. These ensure that core evaluation metrics can still be obtained within the project timeframe.

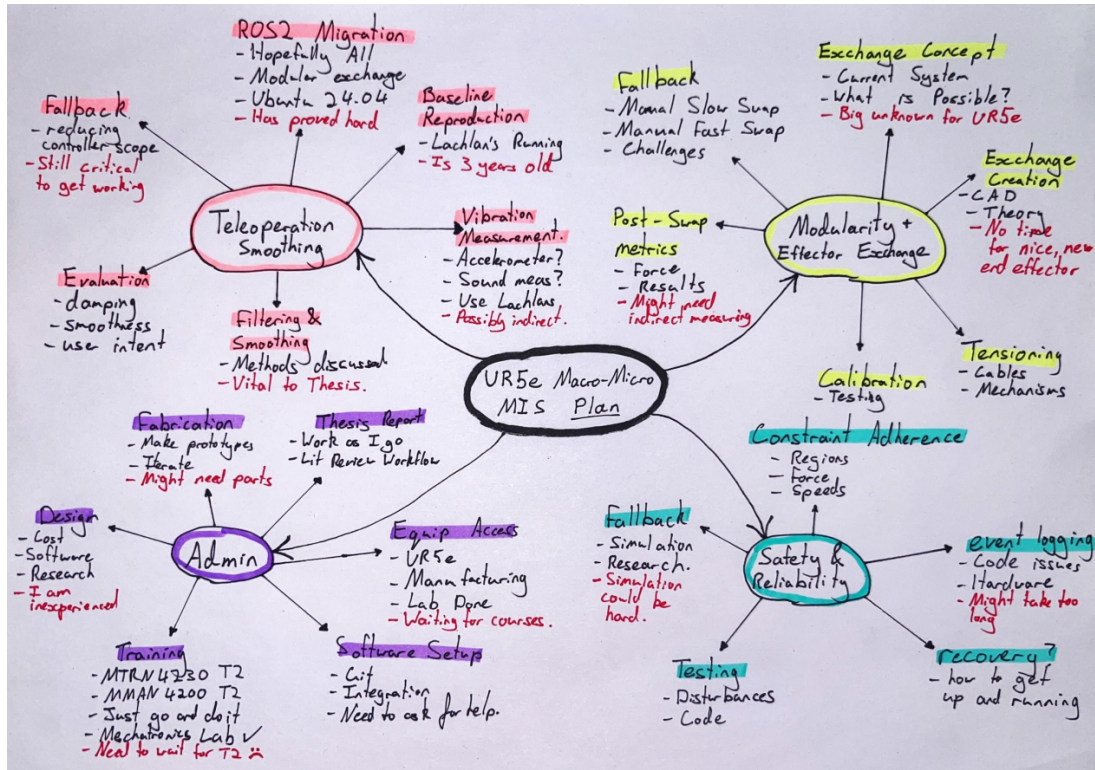


Figure 7: Mind map used to guide project planning, including risks and alternative approaches.

Overall, the project plan is intentionally structured to reflect the narrative developed in the literature review. By focusing on teleoperation smoothing, modular exchange recovery, and safety evaluation within a single system, the work aims to provide a coherent assessment of macro–micro robotic performance under realistic conditions, rather than treating each component in isolation.

4 Project Dependent Preparations

Preparatory work has been undertaken to ensure that the project can proceed without major revision, focusing on software upskilling, system access, and early risk identification. These preparations align with the key challenges identified in the literature, particularly the reliance on robust software integration, reliable teleoperation infrastructure, and repeatable experimental workflows.

4.1 Technical Preparation and Upskilling

A working Ubuntu development environment has been established and ROS2 has been successfully installed and tested. Basic publisher–subscriber communication has been verified using demonstration nodes, confirming correct system setup and message passing

within the ROS2 framework.

This preparation directly addresses a key project dependency, as prior work was implemented in ROS and will require adaptation to ROS2. Early inspection of Scott’s (2022) codebase and system architecture has provided familiarity with the macro–micro teleoperation pipeline, reducing integration uncertainty and establishing a baseline understanding of system structure.

Formal upskilling is also in progress through enrolment in MTRN4230, which provides training in robot control and ROS-based systems and is required for UR5e access. In addition, MMAN4200 (Additive Manufacturing) supports potential iteration of modular end-effector components.

4.2 System and Resource Readiness

Mechatronics laboratory induction has been completed, and the UR5e system and existing continuum end-effector have been physically inspected. This provides practical understanding of the platform and confirms system availability.

Access procedures for the robot have been established, with experimental work to commence following completion of required training. The project also benefits from building on an existing macro–micro system, reducing development risk and allowing focus on performance improvement rather than full system construction.

4.3 Preliminary Observations and Risk Mitigation

Early preparation has identified several key risks consistent with the literature:

- **Software integration:** Migration from ROS to ROS2 presents a non-trivial challenge. This has been partially mitigated through early environment setup and code familiarisation.
- **Hardware access:** Dependence on training and lab availability is addressed through course enrolment and confirmed booking procedures.
- **System complexity:** The coupled nature of macro–micro systems is mitigated by leveraging an existing validated platform rather than developing new hardware.

These preparations demonstrate that the project is ready to proceed, with key technical dependencies established and major risks identified at an early stage.

5 Conclusion

The reviewed literature converges on three key points. Macro–micro and tendon-driven continuum systems are well suited to MIS dexterity, but remain sensitive to vibration, modelling error, and teleoperation instability. Control strategies improve performance, yet are typically validated under constrained conditions. Accessibility is further limited by full-system burden rather than capital cost alone.

The primary gap is therefore not isolated improvements, but integrated validation of system-level performance. In particular, there is limited evidence linking teleoperation smoothness, modular end-effector exchange, and safety-critical reliability within a single, repeatable workflow.

This motivates a systems-level investigation of a UR5e-based macro–micro platform, where control, modularity, and safety are evaluated together under realistic conditions.

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